

Chapter 4 Vector Spaces

Section 4.1: Vector Spaces and Subspaces

Section 4.2: Null Spaces, Column Spaces,
Row Spaces, and Linear Transformation

Section 4.3: Linearly Independent Set; Bases

Vector Spaces

- **Definition:** A **vector space** is a nonempty set V of objects, called *vectors*, on which are defined two operations,
 - **addition**
 - **and multiplication by scalars** (real numbers),
- The axioms must hold for all vectors $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ in V and for all scalars c and d .

Vector Spaces

1. The sum of $\mathbf{u}$ and $\mathbf{v}$, $\mathbf{u} + \mathbf{v}$ is in V .
2. $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$
3. $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$
4. There is a zero vector in V such that $\mathbf{u} + \mathbf{0} = \mathbf{u}$.
5. For each $\mathbf{u}$ in V , there is a vector $-\mathbf{u}$ in V such that $\mathbf{u} + (-\mathbf{u}) = \mathbf{0}$.
6. The scalar multiple of $\mathbf{u}$ by c , $c\mathbf{u}$, is in V .
7. $c(\mathbf{u} + \mathbf{v}) = c\mathbf{u} + c\mathbf{v}$
8. $(c + d)\mathbf{u} = c\mathbf{u} + d\mathbf{u}$
9. $c(d\mathbf{u}) = (cd)\mathbf{u}$
10. $1\mathbf{u} = \mathbf{u}$

Vector Spaces

- For each $\mathbf{u}$ in V and scalar c ,

$$0\mathbf{u} = \mathbf{0}$$

$$c\mathbf{0} = \mathbf{0}$$

$$-\mathbf{u} = (-1)\mathbf{u}$$

Example: The set $\mathbb{R}^n$ is a vector space under the usual addition and scalar multiplication.

The Polynomials of Degree at Most n

Example: For $n \geq 0$, the set $\mathbb{P}_n$ of polynomials of degree at most n , is a vector space.

- It consists of all polynomials of the form

$$\mathbf{p}(t) = a_0 + a_1t + a_2t^2 + \cdots + a_nt^n$$

where the coefficients $a_0, \dots, a_n$ and the variable t are real numbers.

- The **degree** of $\mathbf{p}$ is the highest power of t whose coefficient is not zero. If all the coefficients are zero, $\mathbf{p}$ is called the **zero polynomial**.
- $\mathbb{P}_n$ is a vector space.

Subspaces

- **Definition:** A **subspace** of a vector space V is a subset H of V that has three properties:
 - a. The **zero vector** of V is in H .
 - b. H is closed under **vector addition**. That is, for each $\mathbf{u}$ and $\mathbf{v}$ in H , the sum $\mathbf{u} + \mathbf{v}$ is in H .
 - c. H is closed under **multiplication by scalars**. That is, for each $\mathbf{u}$ in H and each scalar c , the vector $c\mathbf{u}$ is in H .
- Every subspace is a vector space.
- Conversely, every vector space is a subspace (of itself and possibly of other larger spaces).

Subspaces

- The set consisting of only the zero vector in a vector space V is a subspace of V , called the **zero subspace** and written as $\{\mathbf{0}\}$.
- Recall that the term linear combination refers to any sum of scalar multiples of vectors, and

$$\text{Span } \{\mathbf{v}_1, \dots, \mathbf{v}_p\}$$

denotes the set of all vectors that can be written as linear combinations of $\mathbf{v}_1, \dots, \mathbf{v}_p$.

A Subspaces Spanned by a Set

- **Example:** Given $\mathbf{v}_1$ and $\mathbf{v}_2$ in a vector space V , let $H = \text{Span} \{ \mathbf{v}_1, \mathbf{v}_2 \}$.

Show that H is a subspace of V .

- **Solution:**

a. The zero vector is in H , since $\mathbf{0} = 0\mathbf{v}_1 + 0\mathbf{v}_2$.

b. Take two arbitrary vectors in H , say

$\mathbf{w} = t_1\mathbf{v}_1 + t_2\mathbf{v}_2$ and $\mathbf{u} = s_1\mathbf{v}_1 + s_2\mathbf{v}_2$, then

$\mathbf{u} + \mathbf{w} = (s_1 + t_1)\mathbf{v}_1 + (s_2 + t_2)\mathbf{v}_2$ is in H .

c. If c is any scalar, $c\mathbf{u} = (cs_1)\mathbf{v}_1 + (cs_2)\mathbf{v}_2$ is in H .

Thus H is a subspace of V .

A Subspaces Spanned by a Set

- **Theorem:** If $\mathbf{v}_1, \dots, \mathbf{v}_p$ are in a vector space V , then $\text{Span} \{ \mathbf{v}_1, \dots, \mathbf{v}_p \}$ is a subspace of V .
- We call $\text{Span} \{ \mathbf{v}_1, \dots, \mathbf{v}_p \}$ **the subspace spanned** (or **generated**) by $\{ \mathbf{v}_1, \dots, \mathbf{v}_p \}$.
- Give any subspace H of V , a **spanning** (or **generating**) set for H is a set $\{ \mathbf{v}_1, \dots, \mathbf{v}_p \}$ in H such that

$$H = \text{Span} \{ \mathbf{v}_1, \dots, \mathbf{v}_p \}.$$

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Null Space of a Matrix

- **Definition:** The null space of an $m \times n$ matrix A , $\text{Nul } A$, is the set of all solutions of the homogeneous equation $A\mathbf{x} = \mathbf{0}$.

$$\text{Nul } A = \{\mathbf{x}: \mathbf{x} \text{ is in } \mathbb{R}^n \text{ and } A\mathbf{x} = \mathbf{0}\}$$

- $\text{Nul } A$ is a subset of $\mathbb{R}^n$ because A has n columns.
- **Theorem:** The null space of an $m \times n$ matrix A is a subspace of $\mathbb{R}^n$.

Null Space of a Matrix

- **Proof:** We need to show that $\text{Nul } A$ satisfies the three properties of a subspace.
 - a. The vector $\mathbf{0}$ is in $\text{Nul } A$, since $A\mathbf{0} = \mathbf{0}$.
 - b. Let $\mathbf{u}$ and $\mathbf{v}$ represent any two vectors in $\text{Nul } A$. Then $A\mathbf{v} = \mathbf{0}$ and $A\mathbf{u} = \mathbf{0}$, so $A(\mathbf{u} + \mathbf{v}) = A\mathbf{u} + A\mathbf{v} = \mathbf{0}$ showing $\mathbf{u} + \mathbf{v}$ is in $\text{Nul } A$.
 - c. If c is any scalar, then $A(c\mathbf{u}) = c(A\mathbf{u}) = c(\mathbf{0}) = \mathbf{0}$ which shows that $c\mathbf{u}$ is in $\text{Nul } A$.

Thus $\text{Nul } A$ is a subspace of $\mathbb{R}^n$.

Null Space of a Matrix

- There is no obvious relation between vectors in $\text{Nul } A$ and the entries in A .
- We say that $\text{Nul } A$ is defined *implicitly*, because it is defined by a condition that must be checked.

Null Space of a Matrix

- **Example:** Find a spanning set for the null space of the matrix $A\mathbf{x} = \mathbf{0}$

$$A = \begin{bmatrix} -3 & 6 & -1 & 1 & -7 \\ 1 & -2 & 2 & 3 & -1 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix}$$

Null Space of a Matrix

- Solution:

$$\begin{bmatrix} 1 & -2 & 0 & -1 & 3 & 0 \\ 0 & 0 & 1 & 2 & -2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$x_1 - 2x_2 - x_4 + 3x_5 = 0$$

$$x_3 + 2x_4 - 2x_5 = 0$$

$$0 = 0$$

The general solution is

$$x_1 = 2x_2 + x_4 - 3x_5, \quad x_3 = -2x_4 + 2x_5,$$

with x_2 , x_4 , and x_5 free.

Null Space of a Matrix

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 2x_2 + x_4 - 3x_5 \\ x_2 \\ -2x_4 + 2x_5 \\ x_4 \\ x_5 \end{bmatrix} = x_2 \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 1 \\ 0 \\ -2 \\ 1 \\ 0 \end{bmatrix} + x_5 \begin{bmatrix} -3 \\ 0 \\ 2 \\ 0 \\ 1 \end{bmatrix}$$

$\uparrow$ $\uparrow$ $\uparrow$
u **v** **w**

Null Space of a Matrix

- Every linear combination of $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ is an element of $\text{Nul } A$.
- Moreover $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ is a spanning set for $\text{Nul } A$.
- The spanning set produced by this method is automatically linearly independent because the free variables are the weights on the spanning vectors.
- When $\text{Nul } A$ contains nonzero vectors, the number of vectors in the spanning set for $\text{Nul } A$ equals the number of free variables in the equation $A\mathbf{x} = \mathbf{0}$.

Column Space of a Matrix

- **Definition:** The column space of an $m \times n$ matrix A , written as $\text{Col } A$, is the set of all linear combinations of the columns of A . If $A = [\mathbf{a}_1 \ \dots \ \mathbf{a}_n]$, then

$$\text{Col } A = \text{Span}\{\mathbf{a}_1, \dots, \mathbf{a}_n\}$$

- A typical vector in $\text{Col } A$ can be written as $A\mathbf{x}$ for some $\mathbf{x}$, because the $A\mathbf{x}$ is a linear combination of the columns of A . That is,

$$\text{Col } A = \{\mathbf{b}: \mathbf{b} = A\mathbf{x} \text{ for some } \mathbf{x} \text{ in } \mathbb{R}^n\}$$

Column Space of a Matrix

- **Theorem:** The column space of an $m \times n$ matrix A is a subspace of $\mathbb{R}^m$.
- The notation $A\mathbf{x}$ for vectors in $\text{Col } A$ also shows that $\text{Col } A$ is the range of the linear transformation $\mathbf{x} \mapsto A\mathbf{x}$.
- The column space of an $m \times n$ matrix A is all of $\mathbb{R}^m$ if and only if the equation $A\mathbf{x} = \mathbf{b}$ has a solution for every $\mathbf{b}$ in $\mathbb{R}^m$.

Column Space of a Matrix

• **Example:** Let $A = \begin{bmatrix} 2 & 4 & -2 & 1 \\ -2 & -5 & 7 & 3 \\ 3 & 7 & -8 & 6 \end{bmatrix}$ $\mathbf{u} = \begin{bmatrix} 3 \\ -2 \\ -1 \\ 0 \end{bmatrix}$
and $\mathbf{v} = \begin{bmatrix} 3 \\ -1 \\ 3 \end{bmatrix}$.

- Determine if $\mathbf{u}$ is in $\text{Nul } A$. Could $\mathbf{u}$ be in $\text{Col } A$?
- Determine if $\mathbf{v}$ is in $\text{Col } A$. Could $\mathbf{v}$ be in $\text{Nul } A$?

Column Space of a Matrix

- a. An explicit description of $\text{Nul } A$ is not needed here. Simply compute the product $A\mathbf{u}$.

$$A\mathbf{u} = \begin{bmatrix} 2 & 4 & -2 & 1 \\ -2 & -5 & 7 & 3 \\ 3 & 7 & -8 & 6 \end{bmatrix} \begin{bmatrix} 3 \\ -2 \\ -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ -3 \\ 3 \end{bmatrix} \neq \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

The vector $\mathbf{u}$ is *not* a solution of $A\mathbf{x} = \mathbf{0}$, so $\mathbf{u}$ is not in $\text{Nul } A$.

Also, with four entries, $\mathbf{u}$ could not possibly be in $\text{Col } A$, since $\text{Col } A$ is a subspace of $\mathbb{R}^3$.

Column Space of a Matrix

b. Reduce $[A \quad \mathbf{v}]$ to an echelon form.

$$\begin{bmatrix} 2 & 4 & -2 & 1 & 3 \\ -2 & -5 & 7 & 3 & -1 \\ 3 & 7 & -8 & 6 & 3 \end{bmatrix} \sim \begin{bmatrix} 2 & 4 & -2 & 1 & 3 \\ 0 & 1 & -5 & -4 & -2 \\ 0 & 0 & 0 & 17 & 1 \end{bmatrix}$$

The equation $A\mathbf{x} = \mathbf{v}$ is consistent, so $\mathbf{v}$ is in $\text{Col } A$.
With only three entries, $\mathbf{v}$ could not possibly be in $\text{Nul } A$, since $\text{Nul } A$ is a subspace of $\mathbb{R}^4$.

The Row Space

- If A is an $m \times n$ matrix, each row of A has n entries and thus can be identified with a vector in $\mathbb{R}^n$.
- The set of all linear combinations of the row vectors is called the **row space** of A and is denoted by $\text{Row } A$.
- Each row has n entries, so $\text{Row } A$ is a subspace of $\mathbb{R}^n$.
- Since the rows of A are identified with the columns of A^T , we could also write $\text{Col } A^T$ in place of $\text{Row } A$.

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Linearly Independent Set

- An indexed set of vectors $\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ in V is said to be **linearly independent** if the vector equation

$$c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \dots + c_p \mathbf{v}_p = \mathbf{0} \quad (1)$$

has **only** the trivial solution, $c_1 = 0, \dots, c_p = 0$.

- The set $\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ is said to be **linearly dependent** if (1) has a nontrivial solution, *i.e.*, if there are some weights, $c_1, \dots, c_p$, *not all zero*, such that (1) holds.
- In such a case, (1) is called a **linear dependence relation** among $\mathbf{v}_1, \dots, \mathbf{v}_p$.

Linearly Independent Set

- **Theorem:** An indexed set $\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ of two or more vectors, with $\mathbf{v}_1 \neq 0$, is linearly dependent if and only if some $\mathbf{v}_j$ (with $j > 1$) is a linear combination of the preceding vectors, $\mathbf{v}_1, \dots, \mathbf{v}_{j-1}$.
- **Definition:** Let H be a subspace of a vector space V . A set of vectors B in V is a basis for H if
 - (i) B is a linearly independent set, and
 - (ii) $H = \text{Span } B$

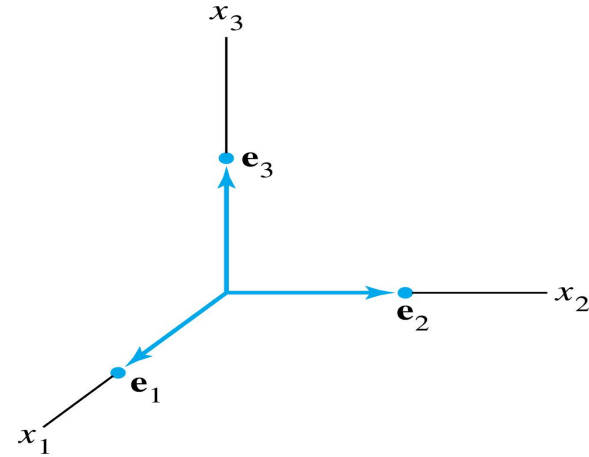
Linearly Independent Set; Bases

- A basis of V is a linearly independent set that spans V .
- The definition of a basis applies to the case when $H = V$, because any vector space is a subspace of itself.
- When $H \neq V$, condition (ii) includes the requirement that each of the vectors $\mathbf{b}_1, \dots, \mathbf{b}_p$ must belong to H , because $\text{Span}\{\mathbf{b}_1, \dots, \mathbf{b}_p\}$ contains $\mathbf{b}_1, \dots, \mathbf{b}_p$.

Standard Basis

- Let $\mathbf{e}_1, \dots, \mathbf{e}_n$ be the columns of the $n \times n$ matrix, I_n .

$$\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix}, \dots, \mathbf{e}_n = \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix}$$



The standard basis for $\mathbb{R}^3$.

- The set $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ is called the **standard basis** for $\mathbb{R}^n$.

The Spanning Set Theorem

- **Theorem:** Let $S = \{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ be a set in V , and let $H = \text{Span}\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$
 - a. If one of the vectors in S —say, $\mathbf{v}_k$ —is a linear combination of the remaining vectors in S , then the set formed from S by removing $\mathbf{v}_k$ still spans H .
 - b. If $H \neq \{\mathbf{0}\}$, some subset of S is a basis for H .

The Spanning Set Theorem

Proof:

a. By rearranging the list of vectors in S , if necessary, we may suppose that $\mathbf{v}_p$ is a linear combination of $\mathbf{v}_1, \dots, \mathbf{v}_{p-1}$

$$\mathbf{v}_p = a_1 \mathbf{v}_1 + \dots + a_{p-1} \mathbf{v}_{p-1}$$

- Given any $\mathbf{x}$ in H , write $\mathbf{x} = c_1 \mathbf{v}_1 + \dots + c_{p-1} \mathbf{v}_{p-1} + c_p \mathbf{v}_p$ for suitable scalars $c_1, \dots, c_p$.

- Substituting the expression for $\mathbf{v}_p$ into the expression for $\mathbf{x}$, it is easy to see that $\mathbf{x}$ is a linear combination of $\mathbf{v}_1, \dots, \mathbf{v}_{p-1}$.

- Thus $\{\mathbf{v}_1, \dots, \mathbf{v}_{p-1}\}$ spans H , because $\mathbf{x}$ was an arbitrary element of H .

The Spanning Set Theorem

- b. If the original spanning set S is linearly independent, then it is already a basis for H .
- Otherwise, one of the vectors in S depends on the others can be deleted.
 - So long as there are two or more vectors in the spanning set, we can repeat this process until the spanning set is linearly independent and hence is a basis for H .
 - If the spanning set is eventually reduced to one vector, that vector will be nonzero (and hence linearly independent) because

$$H \neq \{\mathbf{0}\}$$

The Spanning Set Theorem

- **Example:** Let $\mathbf{v}_1 = \begin{bmatrix} 0 \\ 2 \\ -1 \end{bmatrix}$, $\mathbf{v}_2 = \begin{bmatrix} 2 \\ 2 \\ 0 \end{bmatrix}$, $\mathbf{v}_3 = \begin{bmatrix} 6 \\ 16 \\ -5 \end{bmatrix}$

and $H = \text{Span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$. Note that $\mathbf{v}_3 = 5\mathbf{v}_1 + 3\mathbf{v}_2$.

- Show that $\text{Span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\} = \text{Span}\{\mathbf{v}_1, \mathbf{v}_2\}$, and then find a basis for the subspace H .
- **Solution:** Every vector in $\text{Span}\{\mathbf{v}_1, \mathbf{v}_2\}$ belongs to H because
$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + 0\mathbf{v}_3$$

The Spanning Set Theorem

- Now let $\mathbf{x}$ be any vector in H —say,

$$\mathbf{x} = c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + c_3 \mathbf{v}_3$$

- Since $\mathbf{v}_3 = 5\mathbf{v}_1 + 3\mathbf{v}_2$, we may substitute

$$\mathbf{x} = c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + c_3 (5\mathbf{v}_1 + 3\mathbf{v}_2)$$

$$= (c_1 + 5c_3) \mathbf{v}_1 + (c_2 + 3c_3) \mathbf{v}_2$$

- Thus $\mathbf{x}$ is in $\text{Span} \{\mathbf{v}_1, \mathbf{v}_2\}$, so every vector in H already belongs to $\text{Span} \{\mathbf{v}_1, \mathbf{v}_2\}$.
- We conclude that H and $\text{Span} \{\mathbf{v}_1, \mathbf{v}_2\}$ are actually the same set of vectors.
- It follows that $\{\mathbf{v}_1, \mathbf{v}_2\}$ is a basis of H since $\{\mathbf{v}_1, \mathbf{v}_2\}$ is linearly independent.

Basis for Columns

- **Example:** Find a basis for $\text{Col } B$, where

$$B = [b_1 \quad b_2 \quad \cdots \quad b_5] = \begin{bmatrix} 1 & 4 & 0 & 2 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

- **Solution:** Each nonpivot column of B is a linear combination of the pivot columns.
- In fact, $\mathbf{b}_2 = 4\mathbf{b}_1$ and $\mathbf{b}_4 = 2\mathbf{b}_1 - \mathbf{b}_3$.
- By the Spanning Set Theorem, we may discard $\mathbf{b}_2$ and $\mathbf{b}_4$, and $\{\mathbf{b}_1, \mathbf{b}_3, \mathbf{b}_5\}$ will still span $\text{Col } B$.

Basis for Columns

- Let

$$S = \{b_1, b_3, b_5\} = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} \right\}$$

- Since $b_1 \neq 0$ and no vector in S is a linear combination of the vectors that precede it, S is linearly independent.
- Thus S is a basis for $\text{Col } B$.

Basis for Columns

- **Theorem:** The pivot columns of a matrix A form a basis for $\text{Col } A$.
- **Proof:** Let B be the reduced echelon form of A .
- The set of pivot columns of B is linearly independent, for no vector in the set is a linear combination of the vectors that precede it.
- Since A is row equivalent to B , the pivot columns of A are linearly independent as well, because any linear dependence relation among the columns of A corresponds to a linear dependence relation among the columns of B .

Basis for Columns

- For this reason, every nonpivot column of A is a linear combination of the pivot columns of A .
- Thus the nonpivot columns of A may be discarded from the spanning set for $\text{Col } A$, by the Spanning Set Theorem.
- This leaves the pivot columns of A as a basis for $\text{Col } A$.
- **Warning:** The pivot columns of a matrix A are evident when A has been reduced only to echelon form. But, be careful to use the pivot columns of A itself for the basis of $\text{Col } A$.

The Row Space

- **Theorem:** If two matrices A and B are row equivalent, then their row spaces are the same. If B is in echelon form, the nonzero rows of B form a basis for the row space of A as well as for that of B .
- **Proof:** If B is obtained from A by row operations, the rows of B are linear combinations of the rows of A .
- It follows that any linear combination of the rows of B is automatically a linear combination of the rows of A .

The Row Space

- Thus the row space of B is contained in the row space of A .
- Since row operations are reversible, the same argument shows that the row space of A is a subset of the row space of B .
- So the two row spaces are the same.

The Row Space

- If B is in echelon form, its nonzero rows are linearly independent because no nonzero row is a linear combination of the nonzero rows below it. (Apply Theorem 4 to the nonzero rows of B in reverse order, with the first row last).
- Thus the nonzero rows of B form a basis of the (common) row space of B and A .

Bases for Nul A , Col A , and Row A

- **Example:** Find bases for the row space, the column space, and the null space of the matrix

$$A = \begin{bmatrix} -2 & -5 & 8 & 0 & -17 \\ 1 & 3 & -5 & 1 & 5 \\ 3 & 11 & -19 & 7 & 1 \\ 1 & 7 & -13 & 5 & -3 \end{bmatrix}$$

- **Solution:** To find bases for the row space and the column space, row reduce A to an echelon form:

Bases for Nul A , Col A , and Row A

$$A \sim B = \begin{bmatrix} 1 & 3 & -5 & 1 & 5 \\ 0 & 1 & -2 & 2 & -7 \\ 0 & 0 & 0 & -4 & 20 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

- By Theorem, the first three rows of B form a basis for the row space of A (as well as for the row space of B).
- Thus Basis for Row

$$A: \{(1, 3, -5, 1, 5), (0, 1, -2, 2, -7), (0, 0, 0, -4, 20)\}$$

Bases for Nul A , Col A , and Row A

- For the column space, observe from B that the pivots are in columns 1, 2, and 4.
- Hence columns 1, 2, and 4 of A (not B) form a basis for Col A :

$$\text{Basis for Col } A: \left\{ \begin{bmatrix} -2 \\ 1 \\ 3 \\ 1 \end{bmatrix}, \begin{bmatrix} -5 \\ 3 \\ 11 \\ 7 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 7 \\ 5 \end{bmatrix} \right\}$$

- Notice that any echelon form of A provides (in its nonzero rows) a basis for Row A and also identifies the pivot columns of A for Col A .

Bases for Nul A , Col A , and Row A

- However, for Nul A , we need the *reduced echelon form*.
- Further row operations on B yield

$$A \sim B \sim C = \begin{bmatrix} 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & -2 & 0 & 3 \\ 0 & 0 & 0 & 1 & -5 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Bases for Nul A , Col A , and Row A

- The equation $A\mathbf{x} = \mathbf{0}$ is equivalent to $C\mathbf{x} = \mathbf{0}$, that is,

$$x_1 + x_3 + x_5 = 0$$

$$x_2 - 2x_3 + 3x_5 = 0$$

$$x_4 - 5x_5 = 0$$

- So $x_1 = -x_3 - x_5$, $x_2 = 2x_3 - 3x_5$, $x_4 = 5x_5$, with x_3 and x_5 free variables.

Bases for Nul A , Col A , and Row A

- The calculations show that

$$\text{Basis for Nul } A: \left\{ \begin{bmatrix} -1 \\ 2 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ -3 \\ 0 \\ 5 \\ 1 \end{bmatrix} \right\}$$

- Observe that, unlike the basis for Col A , the bases for Row A and Nul A have no simple connection with the entries in A itself.